

2.1 (a) Complete the entries in the table. Put the sums in the last row. What are the sample means $\bar{x}$ and $\bar{y}$?

x	y	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$	
3	4	2	4	2	4	$\bar{x} = 1$
2	2	1	1	0	0	
1	3	0	0	1	0	$\bar{y} = 2$
1	1	-2	4	-1	2	
0	0	-1	1	-2	2	
$\Sigma x_i = 5$	$\Sigma y_i = 10$	$\Sigma (x_i - \bar{x}) = 0$	$\Sigma (x_i - \bar{x})^2 = 10$	$\Sigma (y_i - \bar{y}) = 0$	$\Sigma (x_i - \bar{x})(y_i - \bar{y}) = 8$	

(b) Calculate b_1 and b_2 using (2.7) and (2.8) and state their interpretation.

$$b_2 = \frac{\Sigma (x_i - \bar{x})(y_i - \bar{y})}{\Sigma (x_i - \bar{x})^2} = \frac{8}{10} = 0.8, \quad b_1 = \bar{y} - b_2 \bar{x} = 2 - 0.8 = 1.2$$

b_1 是指当 $x=0$, fitted line 之截距为 1.2, b_2 是 fitted line 之斜率为 0.8.

(c) Compute $\sum_{i=1}^5 x_i^2$, $\sum_{i=1}^5 x_i y_i$. Using these numerical values, show that $\sum (x_i - \bar{x})^2 = \sum x_i^2 - N\bar{x}^2$ and $\sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - N\bar{x}\bar{y}$.

$$\Sigma_{i=1}^5 x_i^2 = 15, \quad \Sigma_{i=1}^5 x_i y_i = 18, \quad \Sigma_{i=1}^5 x_i^2 - N\bar{x}^2 = 15 - 5 \times 1^2 = 10 = \Sigma_{i=1}^5 (x_i - \bar{x})^2$$

$$\Sigma_{i=1}^5 x_i y_i - N\bar{x}\bar{y} = 18 - 5 \times 2 \times 1 = 18 - 10 = 8 = \Sigma_{i=1}^5 (x_i - \bar{x})(y_i - \bar{y})$$

(d) Use the least squares estimates from part (b) to compute the fitted values of y , and complete the remainder of the table below. Put the sums in the last row.

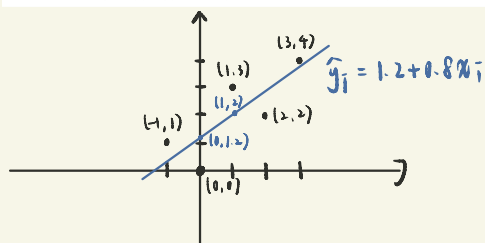
Calculate the sample variance of y , $s_y^2 = \sum_{i=1}^N (y_i - \bar{y})^2 / (N - 1)$, the sample variance of x , $s_x^2 = \sum_{i=1}^N (x_i - \bar{x})^2 / (N - 1)$, the sample covariance between x and y , $s_{xy} = \sum_{i=1}^N (y_i - \bar{y})(x_i - \bar{x}) / (N - 1)$, the sample correlation between x and y , $r_{xy} = s_{xy} / (s_x s_y)$ and the coefficient of variation of x , $CV_x = 100(s_x / \bar{x})$. What is the median, 50th percentile, of x ?

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$	
3	4	3.6	0.4	0.16	1.2	$s_y^2 = \Sigma_{i=1}^5 \frac{(y_i - \bar{y})^2}{N-1} = \frac{4+1+1+4}{5-1} = 2.5$
2	2	2.8	-0.8	0.64	-1.6	
1	3	2	1	1	1	$s_x^2 = \Sigma_{i=1}^5 \frac{(x_i - \bar{x})^2}{N-1} = \frac{10}{4} = 2.5$
1	1	0.4	0.6	0.36	-0.6	
0	0	1.2	-1.2	1.44	0	$s_{xy} = \Sigma_{i=1}^5 (y_i - \bar{y})(x_i - \bar{x}) / (N-1) = \frac{8}{4} = 2$
$\Sigma x_i = 5$	$\Sigma y_i = 10$	$\Sigma \hat{y}_i = 10$	$\Sigma \hat{e}_i = 0$	$\Sigma \hat{e}_i^2 = 3.6$	$\Sigma x_i \hat{e}_i = 0$	$r_{xy} = s_{xy} / (s_x s_y) = 2 / 2.5 = \frac{4}{5} = 0.8$

$$CV_x = 100(s_x / \bar{x}) = 100\sqrt{2.5} = 158.11388.$$

$$\text{median}(x) = 1$$

(e) On graph paper, plot the data points and sketch the fitted regression line $\hat{y}_i = b_1 + b_2 x_i$.



$$\hat{y}_i = 1.2 + 0.8x_i$$

$$x=0 \Rightarrow \hat{y}_i = 1.2, \quad x=1 \Rightarrow \hat{y}_i = 2$$

(f) On the sketch in part (e), locate the point of the means $(\bar{x}, \bar{y})$. Does your fitted line pass through that point? If not, go back to the drawing board, literally.

由 (e) 可知, $(\bar{x}, \bar{y}) = (1, 2)$ 在 fitted line 上.

1g) Show that for these numerical values $\bar{y} = b_1 + b_2\bar{x}$.

$\bar{x} = 1$ fit in fitted line $\Rightarrow 1.2 + 0.8 = 2 = \bar{y} \Rightarrow$ on the fitted line.

1h) Show that for these numerical values $\hat{\bar{y}} = \bar{y}$, where $\hat{\bar{y}} = \sum \hat{y}_i / N$.

$$\hat{\bar{y}} = \frac{\sum \hat{y}_i}{N} = \frac{10}{5} = 2 = \bar{y}$$

1i) Compute $\hat{\sigma}^2$. $\hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{N-2} = \frac{3.6}{5-2} = 1.2$

1j) Compute $\widehat{\text{var}}(b_2|x)$ and $\text{se}(b_2)$.

$$\widehat{\text{var}}(b_2|x) = \frac{\hat{\sigma}^2}{\sum (x_i - \bar{x})^2} = \frac{1.2}{10} = 0.12, \quad \text{se}(b_2) = \sqrt{\widehat{\text{var}}(b_2|x)} = \sqrt{0.12} = 0.3464$$

2.22 A longitudinal experiment was conducted in Tennessee beginning in 1985 and ending in 1989. A single cohort of students was followed from kindergarten through third grade. In the experiment children were randomly assigned within schools into three types of classes: small classes with 13–17 students, regular-sized classes with 22–25 students, and regular-sized classes with a full-time teacher aide to assist the teacher. Student scores on achievement tests were recorded as well as some information about the students, teachers, and schools. Data for the kindergarten classes are contained in the data file *star5_small*. [Note: The data file *star5* contains more observations and variables.]

(a) Using children who are in either a regular-sized class or a small class, estimate the regression model explaining students' combined aptitude scores as a function of class size, $\text{TOTALSCORE}_i = \beta_1 + \beta_2 \text{SMALL}_i + e_i$. Interpret the estimates. Based on this regression result, what do you conclude about the effect of class size on learning?

```
Call:
lm(formula = totalscore ~ small, data = star_a)

Residuals:
    Min       1Q   Median       3Q      Max
-189.44  -50.44   -9.44   43.06   324.38

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  916.442     3.675  249.396  <2e-16 ***
small        12.175     5.369    2.268  0.0236 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 74.59 on 773 degrees of freedom
Multiple R-squared:  0.006608, Adjusted R-squared:  0.005323
F-statistic: 5.142 on 1 and 773 DF, p-value: 0.02363
```

$\hat{\beta}_1 = 916.442$: The students' combined aptitude scores is 916.442 in the regular-sized class.

$\hat{\beta}_2 = 12.175$: Students in small classes scored, on average, 12.175 points higher than those in regular-sized classes. This effect is statistically significant.

In conclusion, the small class can improve learning performance effectively.

(b) Repeat part (a) using dependent variables *READSCORE* and *MATHSCORE*. Do you observe any differences?

```
Call:
lm(formula = readscore ~ small, data = star_a)

Residuals:
    Min       1Q   Median       3Q      Max
-74.665 -21.590  -2.665  16.410  194.335

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  432.665     1.488  290.760  < 2e-16 ***
small         6.924     2.174    3.185  0.00151 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 30.2 on 773 degrees of freedom
Multiple R-squared:  0.01295, Adjusted R-squared:  0.01167
F-statistic: 10.14 on 1 and 773 DF, p-value: 0.001507
```

$$\hat{\beta}_1 = 432.665 \quad \hat{\beta}_2 = 6.924$$

According to the READSCORE results, the average total score is 432.665 in regular-sized class, and small class score 6.924 points higher, which is statistically significant.

In conclusion, small-class teaching has a statistically significant positive impact on reading scores compared to math scores.

```
Call:
lm(formula = mathscore ~ small, data = star_a)

Residuals:
    Min       1Q   Median       3Q      Max
-144.777  -35.028  -5.028   29.223  142.223

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  483.777     2.449  197.545  <2e-16 ***
small         5.251     3.578    1.467   0.143
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 49.71 on 773 degrees of freedom
Multiple R-squared:  0.002778, Adjusted R-squared:  0.001488
F-statistic: 2.153 on 1 and 773 DF, p-value: 0.1427
```

$$\hat{\beta}_1 = 483.777 \quad \hat{\beta}_2 = 5.251$$

According to the MATHSCORE results, the average total score is 483.777 in regular-sized class, and small class score 5.251 points higher, which is not statistically significant.

(c)

Using children who are in either a regular-sized class or a regular-sized class with a teacher aide, estimate the regression model explaining students' combined aptitude scores as a function of the presence of a teacher aide, $TOTALSCORE = \gamma_1 + \gamma_2 AIDE + e$. Interpret the estimates. Based on this regression result, what do you conclude about the effect on learning of adding a teacher aide to the classroom?

```
Call:
lm(formula = totalscore ~ aide, data = star_c)

Residuals:
    Min       1Q   Median       3Q      Max
-191.748  -50.442   -5.442   40.558   312.558

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  916.442     3.559  257.529  <2e-16 ***
aide         4.306       4.994   0.862   0.389
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 72.23 on 835 degrees of freedom
Multiple R-squared:  0.0008898, Adjusted R-squared:  -0.0003068
F-statistic: 0.7436 on 1 and 835 DF, p-value: 0.3888
```

$$\hat{\beta}_1 = 916.442, \hat{\beta}_2 = 4.306 \text{ (not significant)}$$

We found that regular-sized classes with a teacher aide scored 4.306 points higher than those without an aide. However, this difference is not statistically significant.

The small class has better learning performance than the teacher aide.

(d)

Repeat part (c) using dependent variables *READSCORE* and *MATHSCORE*. Do you observe any differences?

```
Call:
lm(formula = readscore ~ aide, data = star_c)

Residuals:
    Min       1Q   Median       3Q      Max
 -74.665  -21.536   -2.536   15.464   194.335

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  432.665     1.460  296.341  <2e-16 ***
aide         2.871       2.049   1.401   0.161
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 29.64 on 835 degrees of freedom
Multiple R-squared:  0.002347, Adjusted R-squared:  0.001152
F-statistic: 1.964 on 1 and 835 DF, p-value: 0.1615
```

$$\hat{\beta}_1 = 432.665, \hat{\beta}_2 = 2.871 \text{ (not significant)}$$

```
Call:
lm(formula = mathscore ~ aide, data = star_c)

Residuals:
    Min       1Q   Median       3Q      Max
-146.212  -31.212   -5.777   29.223   142.223

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  483.777     2.355  205.464  <2e-16 ***
aide         1.435       3.304   0.434   0.664
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 47.79 on 835 degrees of freedom
Multiple R-squared:  0.0002258, Adjusted R-squared:  -0.0009715
F-statistic: 0.1886 on 1 and 835 DF, p-value: 0.6642
```

$$\hat{\beta}_1 = 483.777, \hat{\beta}_2 = 1.435 \text{ (not significant)}$$

While the presence of a teacher aide is associated with slightly higher scores in both reading (+2.871) and math (+1.435), neither effect is statistically significant. This suggests that adding a teacher aide to a regular-sized class is less effective than reducing the class size.

In conclusion, the small class with a better learning performance than the teacher aide whether the subject or sized in class.

2.22 code

```
setwd("G:/我的雲端硬碟/交大/碩一下/econometric/PoE5data")
load("star5_small.rdata")
```

(a)

```
star_a <- subset(star5_small, aide == 0)
model_a <- lm(totalscore ~ small, data = star_a)
summary(model_a)
```

(b) read vs math

```
summary(lm(readscore ~ small, data = star_a))
summary(lm(mathscore ~ small, data = star_a))
```

(c) 助教

```
star_c <- subset(star5_small, small == 0)
model_c <- lm(totalscore ~ aide, data = star_c)
summary(model_c)
```

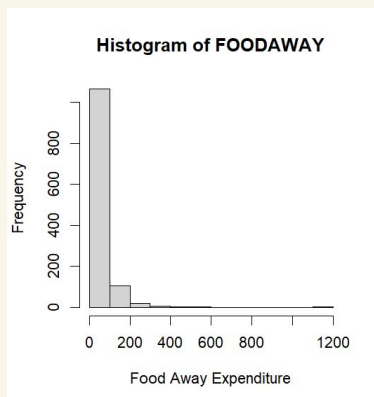
(d)

```
summary(lm(readscore ~ aide, data = star_c))
summary(lm(mathscore ~ aide, data = star_c))
```


2.25

Consumer expenditure data from 2013 are contained in the file *cex5_small*. [Note: *cex5* is a larger version with more observations and variables.] Data are on three-person households consisting of a husband and wife, plus one other member, with incomes between \$1000 per month to \$20,000 per month. *FOODAWAY* is past quarter's food away from home expenditure per month per person, in dollars, and *INCOME* is household monthly income during past year, in \$100 units.

- (a) Construct a histogram of *FOODAWAY* and its summary statistics. What are the mean and median values? What are the 25th and 75th percentiles?



```
> summary(cex5_small$foodaway)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.00   12.04   32.55   49.27   67.50  1179.00
```

Mean is 49.27 and median is 32.55.

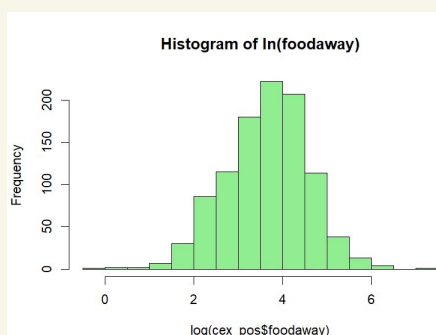
25th and 75th percentiles are 12.04 and 67.5.

- (b) What are the mean and median values of *FOODAWAY* for households including a member with an advanced degree? With a college degree member? With no advanced or college degree member?

```
> aggregate(foodaway ~ college, data = cex5_small, mean)
  college foodaway
1      0 49.56999
2      1 48.59718
> aggregate(foodaway ~ college, data = cex5_small, median)
  college foodaway
1      0  28.89
2      1  36.11
> aggregate(foodaway ~ advanced, data = cex5_small, mean)
  advanced foodaway
1      0 42.76161
2      1 73.15494
> aggregate(foodaway ~ advanced, data = cex5_small, median)
  advanced foodaway
1      0  28.89
2      1  48.15
> no_degree_data <- subset(cex5_small, advanced == 0 & college == 0)
> summary(no_degree_data$foodaway)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.00   9.63   26.02   39.01   52.65   437.78
```

	median	mean
College	36.11	48.597
advanced degree	48.15	73.15
no advanced or college	26.02	39.01

- (c) Construct a histogram of $\ln(\text{FOODAWAY})$ and its summary statistics. Explain why *FOODAWAY* and $\ln(\text{FOODAWAY})$ have different numbers of observations.



```
> summary(log(cex5_small$foodaway))
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  -Inf   2.488   3.483   -Inf   4.212   7.072
```

foodaway: 1200 observations

$\ln(\text{foodaway})$: 1022 observations

The number of observations for $\ln(\text{foodaway})$ is fewer than foodaway because the log-transformation excludes any observations with a value of zero, as $\ln(0)$ is undefined.

- (d) Estimate the linear regression $\ln(\text{FOODAWAY}) = \beta_1 + \beta_2 \text{INCOME} + e$. Interpret the estimated slope.

```
Call:
lm(formula = log(foodaway) ~ income, data = cex_pos)

Residuals:
    Min       1Q   Median       3Q      Max
-3.6547 -0.5777  0.0530  0.5937  2.7000

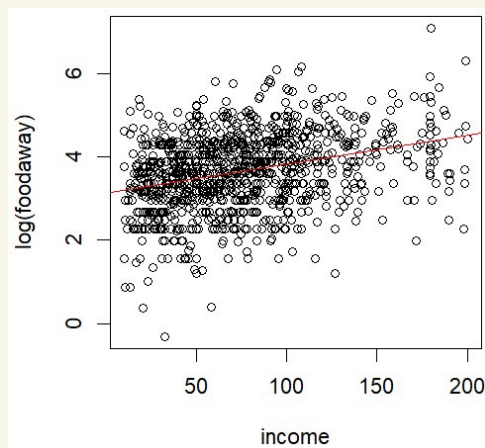
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.1293004   0.0565503   55.34  <2e-16 ***
income       0.0069017   0.0006546   10.54  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8761 on 1020 degrees of freedom
Multiple R-squared:  0.09826,    Adjusted R-squared:  0.09738
F-statistic: 111.1 on 1 and 1020 DF,  p-value: < 2.2e-16
```

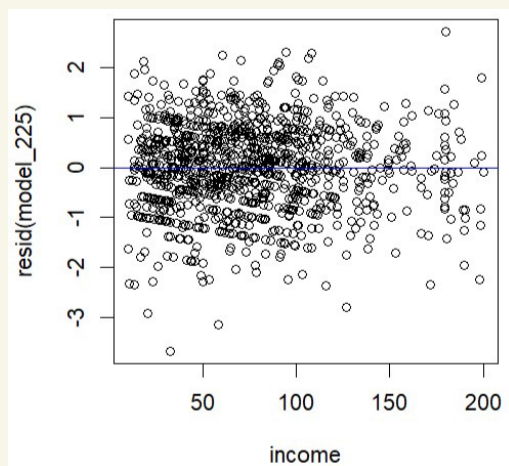
slope: 0.0069017

When income increases by 1 unit (\$100), the food expenditure increases 0.69% on average.

(e) Plot $\ln(\text{FOODAWAY})$ against INCOME , and include the fitted line from part (d).



(f) Calculate the least squares residuals from the estimation in part (d). Plot them vs. INCOME . Do you find any unusual patterns, or do they seem completely random?



Because the residual plot shows that the residuals are randomly scattered around the zero baseline without any discernible pattern or systematic trend, it indicates that the residuals are randomly distributed and the model is appropriately specified.

```
# 2.25 code
setwd("G:/我的雲端硬碟/交大/碩一下/econometric/PoE5data")
load("cex5_small.rdata")

# (a)
hist(cex5_small$foodaway, main="Histogram of FOODAWAY", xlab="Food Away Expenditure")
summary(cex5_small$foodaway)

# (b)
aggregate(foodaway ~ college, data = cex5_small, mean)
aggregate(foodaway ~ college, data = cex5_small, median)

aggregate(foodaway ~ advanced, data = cex5_small, mean)
aggregate(foodaway ~ advanced, data = cex5_small, median)

no_degree_data <- subset(cex5_small, advanced == 0 & college == 0)
summary(no_degree_data$foodaway)

# (c)
hist(log(cex5_small$foodaway), main="Histogram of ln(foodaway)", col="lightgreen")
summary(log(cex5_small$foodaway))
sum(cex5_small$foodaway == 0) # 被剔除掉的人數 (即等於 0 的人數)
sum(cex5_small$foodaway > 0) # 取對數後「剩下」的有效樣本數 (即大於 0 的人數)

# (d)
cex_pos <- subset(cex5_small, foodaway > 0)
model_225 <- lm(log(foodaway) ~ income, data = cex_pos)
summary(model_225)

# (e)
plot(log(foodaway) ~ income, data = cex_pos)
abline(model_225, col="red")

# (f)
plot(resid(model_225) ~ income, data = cex_pos)
abline(h=0, col="blue")
```